Important: Text generated by AI and may contain incorrect information, always check the content.

```python
# Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Check the first few rows of the dataframe
display('First few rows of the DataFrame:')
display(df.head())

# Check the summary statistics for the dataframe
display('Summary statistics for the DataFrame:')
display(df.describe())

# Check the info of the dataframe
display('Info of the DataFrame:')
display(df.info())

# Check for missing values
display('Display of missing values:')
display(df.isnull().sum())

# Check the distribution of categorical variables
categorical_cols = ['GENDER', 'LUNG_CANCER']
display('Distribution of categorical variables:')
for col in categorical_cols:
    display(f'Distribution of {col}:')
    display(df[col].value_counts())

# Check the distribution of numerical variables
numerical_cols = ['AGE', 'SMOKING', 'YELLOW_FINGERS', 'ANXIETY', 'PEER_PRESSURE', 'CHRONIC_DISEASE',
                 'FATIGUE', 'ALLERGY', 'WHEEZING', 'ALCOHOL_CONSUMING', 'COUGHING', 'SHORTNESS_OF_BREATH',
                 'SWALLOWING_DIFFICULTY', 'CHEST_PAIN']
display('Summary statistics for numerical columns:')
display(df[numerical_cols].describe())

# Plot histograms for numerical variables
display('Histograms for numerical variables:')
for col in numerical_cols:
    plt.figure(figsize=(8,6))
    sns.histplot(df[col], kde=True)
    plt.title(f'Histogram of {col}')
    plt.show()

# Check the correlation between numerical variables
numerical_df = df[numerical_cols]
display('Correlation between numerical variables:')
plt.figure(figsize=(10,8))
sns.heatmap(numerical_df.corr(), annot=True, cmap='coolwarm', square=True)
plt.show()

# Check the relationship between categorical and numerical variables
for col in categorical_cols:
    for num_col in numerical_cols:
        plt.figure(figsize=(8,6))
        sns.boxplot(x=col, y=num_col, data=df)
        plt.title(f'Boxplot of {num_col} by {col}')
        plt.show()
```

'First few rows of the DataFrame:'

	GENDER	AGE	SMOKING	YELLOW_FINGERS	ANXIETY	PEER_PRESSURE	CHRONIC_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL_CONSUMING	COUGHING	SHORTNESS_OF_BREATH	SWALLOWING_DIFFICULTY	CHEST_PAIN	LUNG_CANCER
0	M	65	1	1	1	2	2	1	2	2	2	2	2	2	1	NO
1	F	55	1	2	2	1	1	2	2	2	1	1	1	2	2	NO
2	F	78	2	2	1	1	1	2	1	2	1	1	2	1	1	YES
3	M	60	2	1	1	1	2	1	2	1	1	2	1	2	2	YES
4	F	80	1	1	2	1	1	2	1	2	1	1	1	1	2	NO

'Summary statistics for the DataFrame:'

	AGE	SMOKING	YELLOW_FINGERS	ANXIETY	PEER_PRESSURE	CHRONIC_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL_CONSUMING	COUGHING	SHORTNESS_OF_BREATH	SWALLOWING_DIFF
count	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.0
mean	55.169000	1.491000	1.514000	1.494000	1.499000	1.509667	1.489667	1.506667	1.497333	1.491333	1.510667	1.488000	1.4
std	14.723746	0.500002	0.499887	0.500047	0.500082	0.499990	0.499977	0.500039	0.500076	0.500008	0.499970	0.499939	0.4
min	30.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.0
25%	42.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.0
50%	55.000000	1.000000	2.000000	1.000000	1.000000	2.000000	1.000000	2.000000	1.000000	1.000000	2.000000	1.000000	1.0
75%	68.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.0
max	80.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.0

```
'Info of the DataFrame:'
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3000 entries, 0 to 2999
Data columns (total 16 columns):
 #   Column                 Non-Null Count  Dtype 
---  ------                 --------------  ----- 
 0   GENDER                 3000 non-null   object
 1   AGE                    3000 non-null   int64 
 2   SMOKING                3000 non-null   int64 
 3   YELLOW_FINGERS         3000 non-null   int64 
 4   ANXIETY                3000 non-null   int64 
 5   PEER_PRESSURE          3000 non-null   int64 
 6   CHRONIC_DISEASE        3000 non-null   int64 
 7   FATIGUE                3000 non-null   int64 
 8   ALLERGY                3000 non-null   int64 
 9   WHEEZING               3000 non-null   int64 
 10  ALCOHOL_CONSUMING      3000 non-null   int64 
 11  COUGHING               3000 non-null   int64 
 12  SHORTNESS_OF_BREATH    3000 non-null   int64 
 13  SWALLOWING_DIFFICULTY  3000 non-null   int64 
 14  CHEST_PAIN             3000 non-null   int64 
 15  LUNG_CANCER            3000 non-null   object
dtypes: int64(14), object(2)
memory usage: 375.1+ KB
None
'Display of missing values:'
```

	0
GENDER	0
AGE	0

	0
SMOKING	0
YELLOW_FINGERS	0
ANXIETY	0
PEER_PRESSURE	0
CHRONIC_DISEASE	0
FATIGUE	0
ALLERGY	0
WHEEZING	0
ALCOHOL_CONSUMING	0
COUGHING	0
SHORTNESS_OF_BREATH	0
SWALLOWING_DIFFICULTY	0
CHEST_PAIN	0
LUNG_CANCER	0

dtype: int64

```
'Distribution of categorical variables:'
'Distribution of GENDER:'
```

	GENDER
M	1514
F	1486

dtype: int64

```
'Distribution of LUNG_CANCER:'
```

	LUNG_CANCER
YES	1518
NO	1482

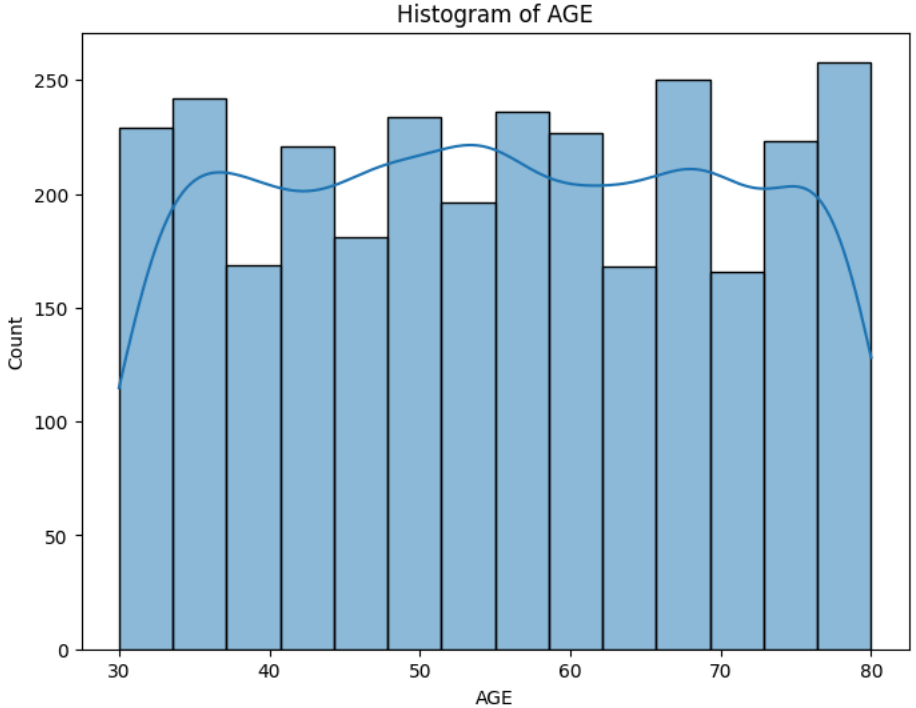
dtype: int64

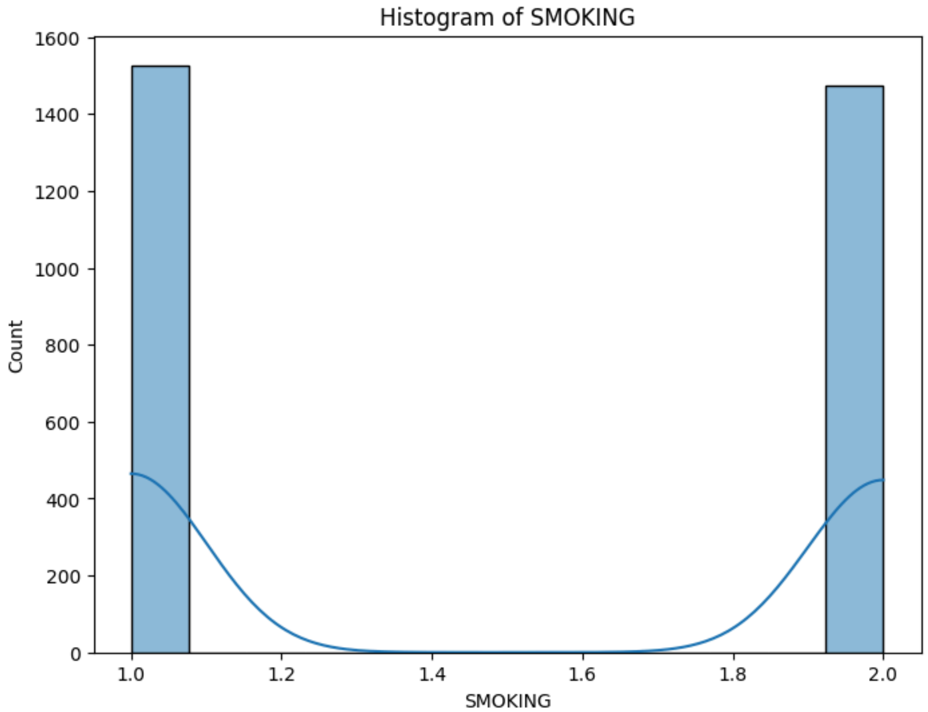
```
'Summary statistics for numerical columns:'
```

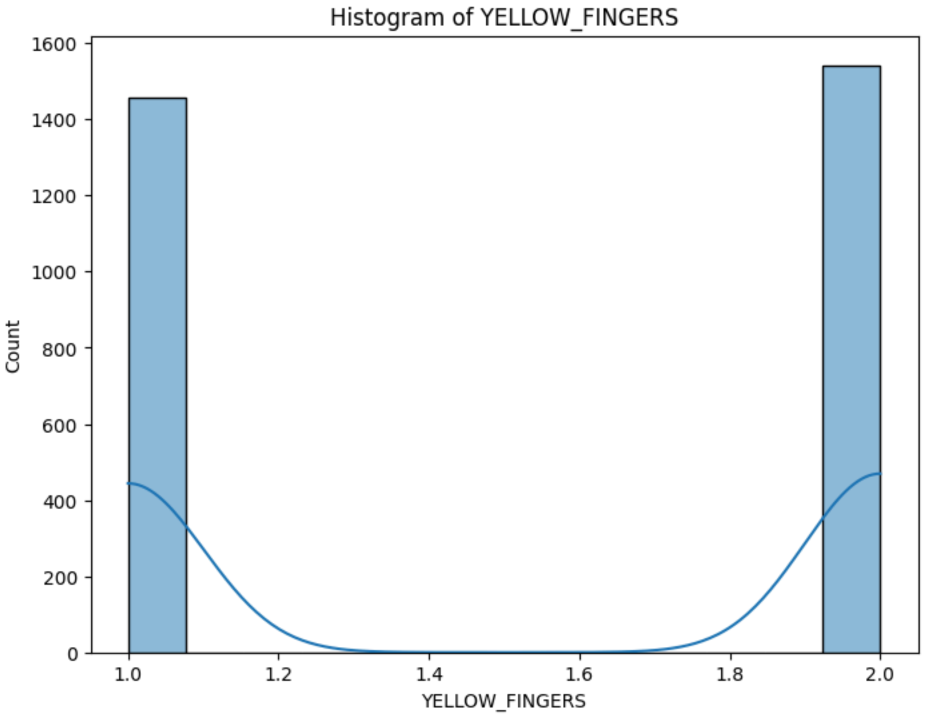
	AGE	SMOKING	YELLOW_FINGERS	ANXIETY	PEER_PRESSURE	CHRONIC_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL_CONSUMING	COUGHING	SHORTNESS_OF_BREATH	SWALLOWING_DIFF
count	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000	3000.0
mean	55.169000	1.491000	1.514000	1.494000	1.499000	1.509667	1.489667	1.506667	1.497333	1.491333	1.510667	1.488000	1.4
std	14.723746	0.500002	0.499887	0.500047	0.500082	0.499990	0.499977	0.500039	0.500076	0.500008	0.499970	0.499939	0.4
min	30.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.0
25%	42.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.0

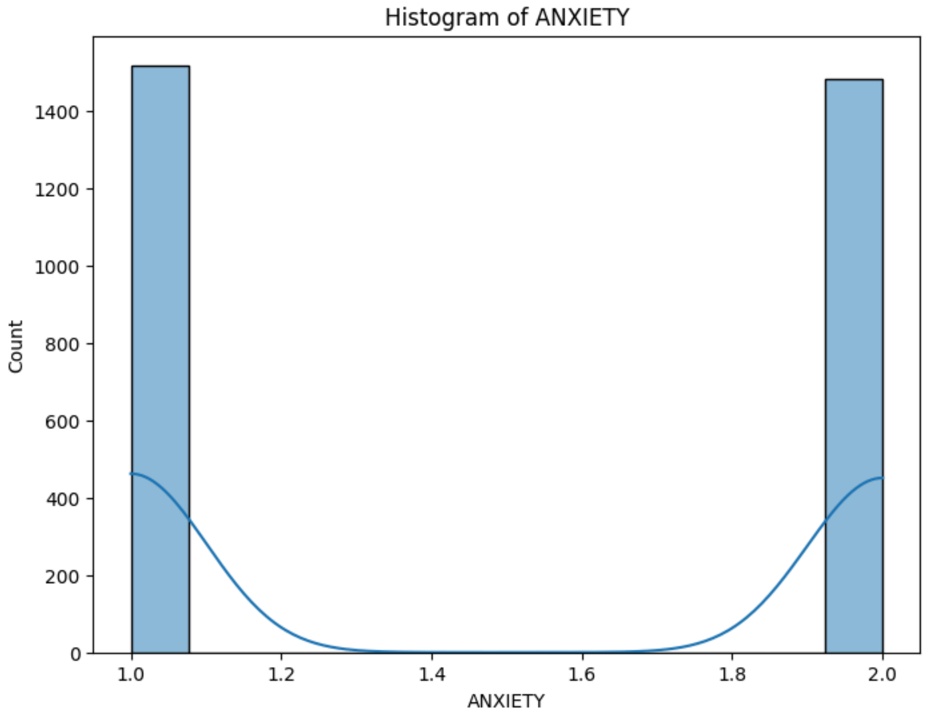
	AGE	SMOKING	YELLOW_FINGERS	ANXIETY	PEER_PRESSURE	CHRONIC_DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL_CONSUMING	COUGHING	SHORTNESS_OF_BREATH	SWALLOWING_DIFF
50%	55.000000	1.000000	2.000000	1.000000	1.000000	2.000000	1.000000	2.000000	1.000000	1.000000	2.000000	1.000000	1.0
75%	68.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.0
max	80.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.0

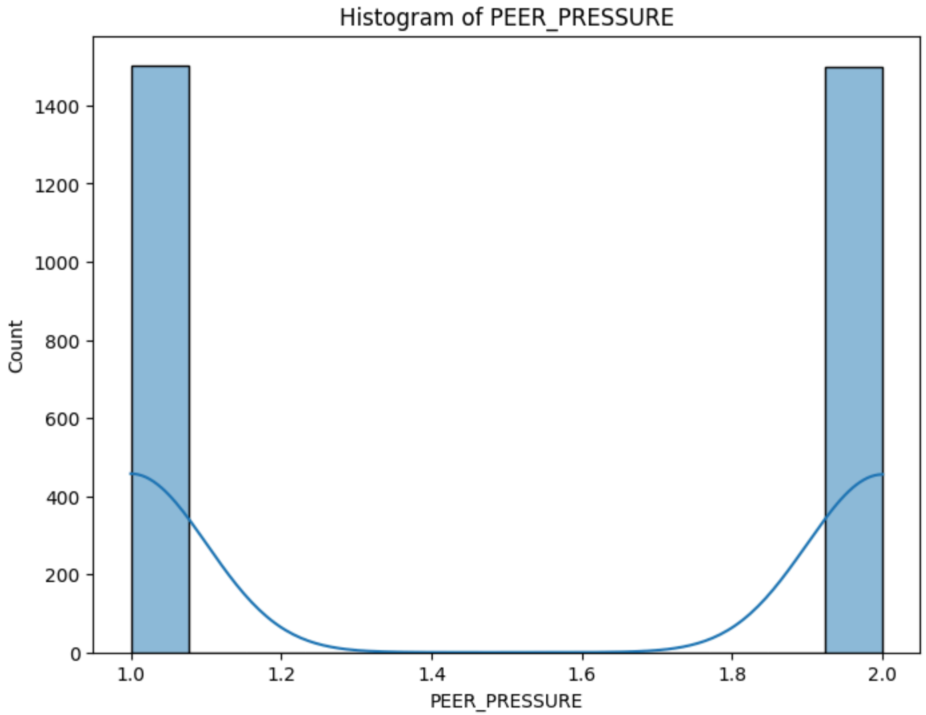
'Histograms for numerical variables:'

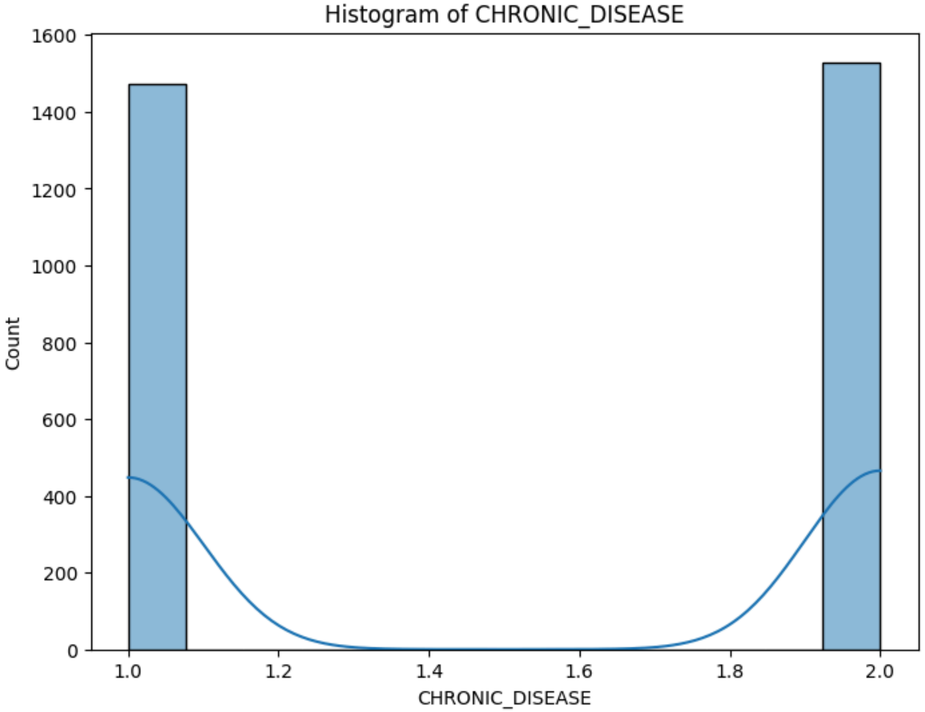


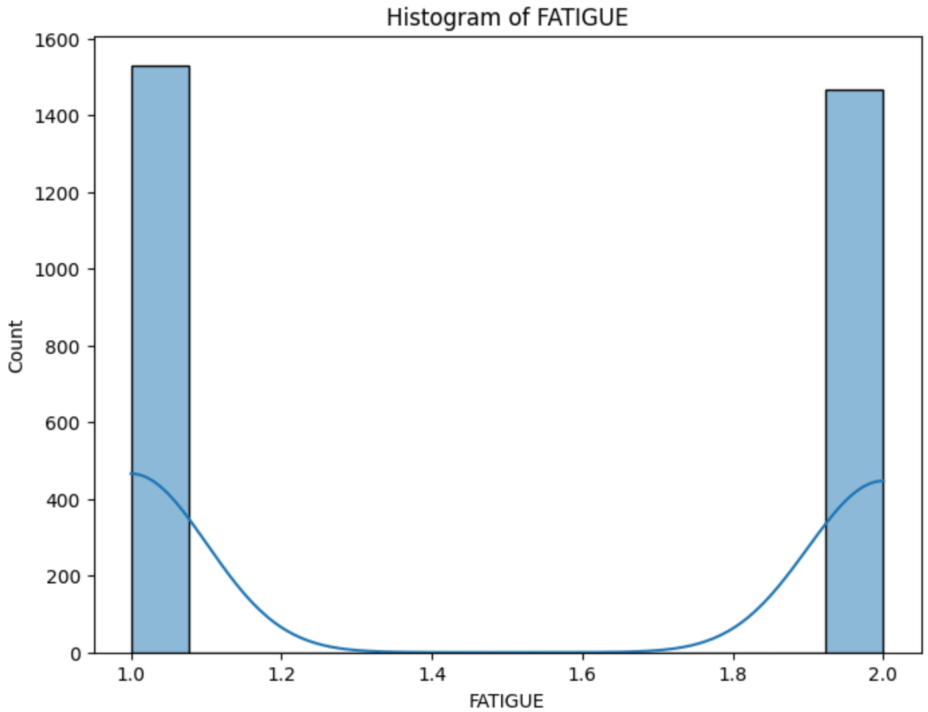

Histogram of SMOKING
1600
1400
1200
1000
800
600
400
200
0
Count
1.0
1.2
1.4
1.6
1.8
2.0
SMOKING


Histogram of YELLOW_FINGERS
Count
YELLOW_FINGERS
0
200
400
600
800
1000
1200
1400
1600
1.0
1.2
1.4
1.6
1.8
2.0

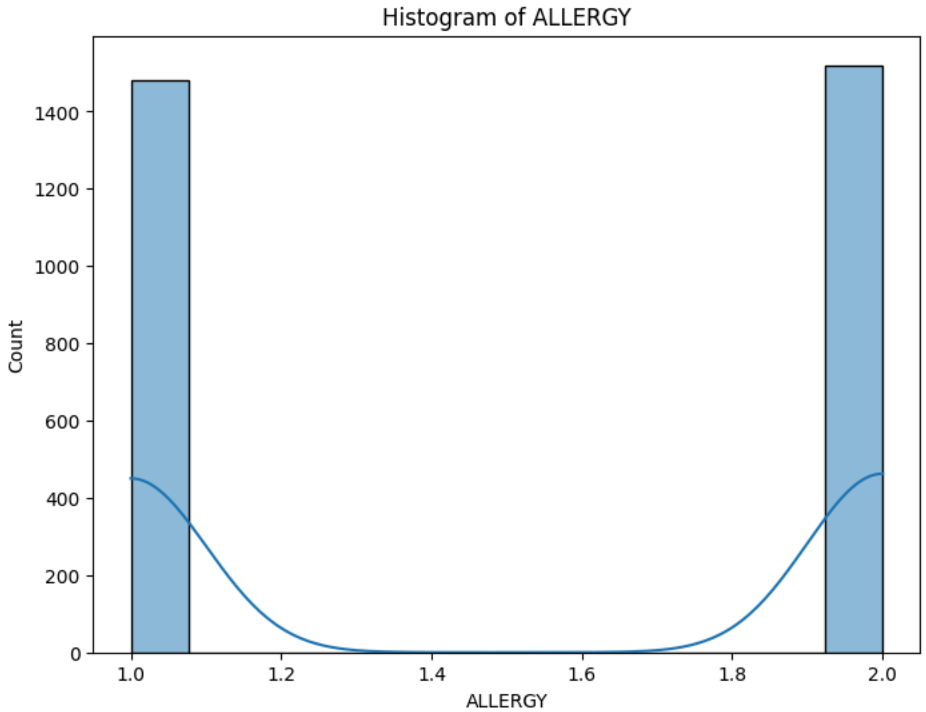

Histogram of ANXIETY
Count
ANXIETY
0
200
400
600
800
1000
1200
1400
1.0
1.2
1.4
1.6
1.8
2.0

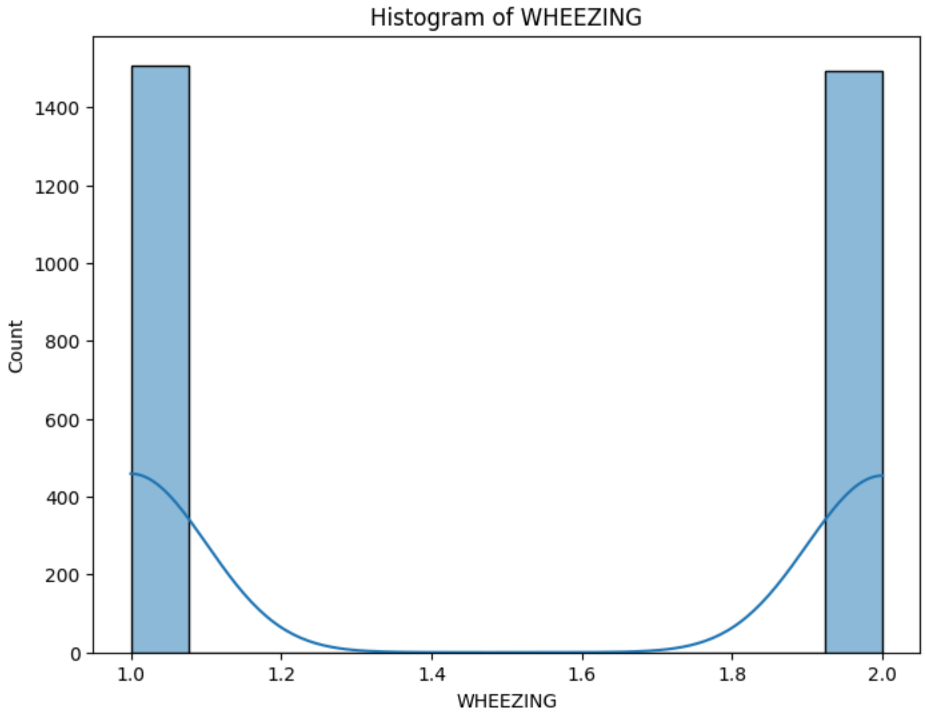

Histogram of PEER_PRESSURE
Count
PEER_PRESSURE
0
200
400
600
800
1000
1200
1400
1.0
1.2
1.4
1.6
1.8
2.0

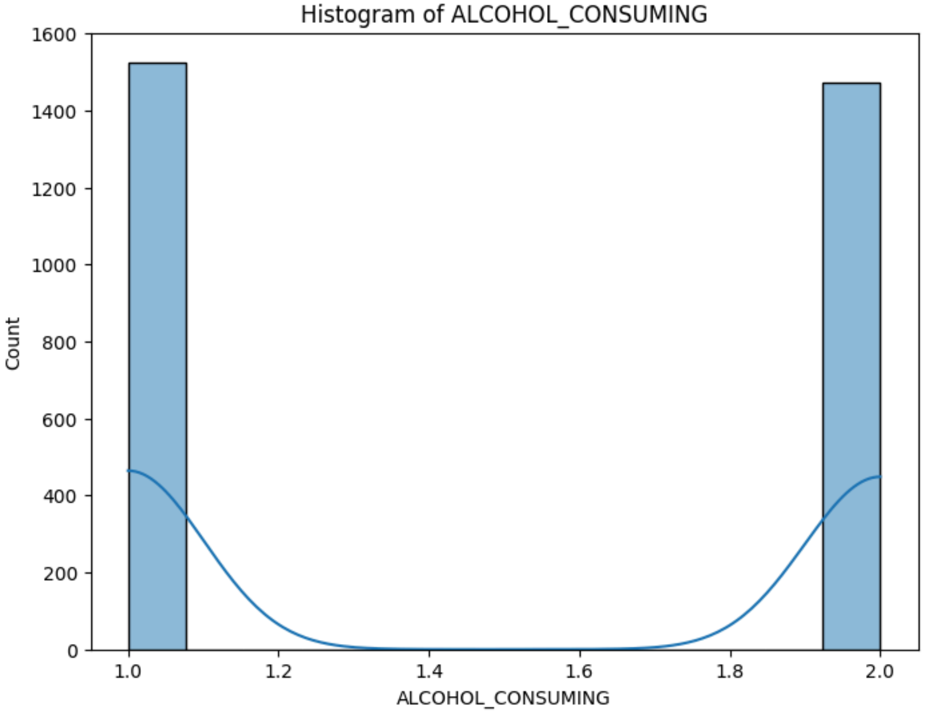

Histogram of CHRONIC_DISEASE
Count
CHRONIC_DISEASE
0
200
400
600
800
1000
1200
1400
1600
1.0
1.2
1.4
1.6
1.8
2.0



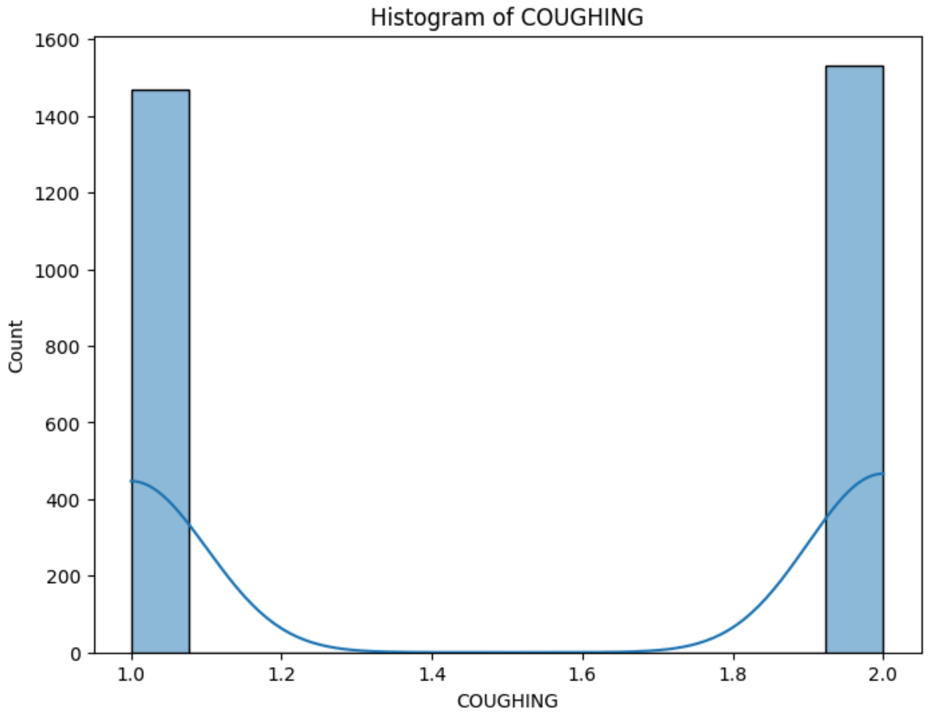
Histogram of FATIGUE
1600
1400
1200
1000
800
600
400
200
0
Count
1.0
1.2
1.4
1.6
1.8
2.0
FATIGUE

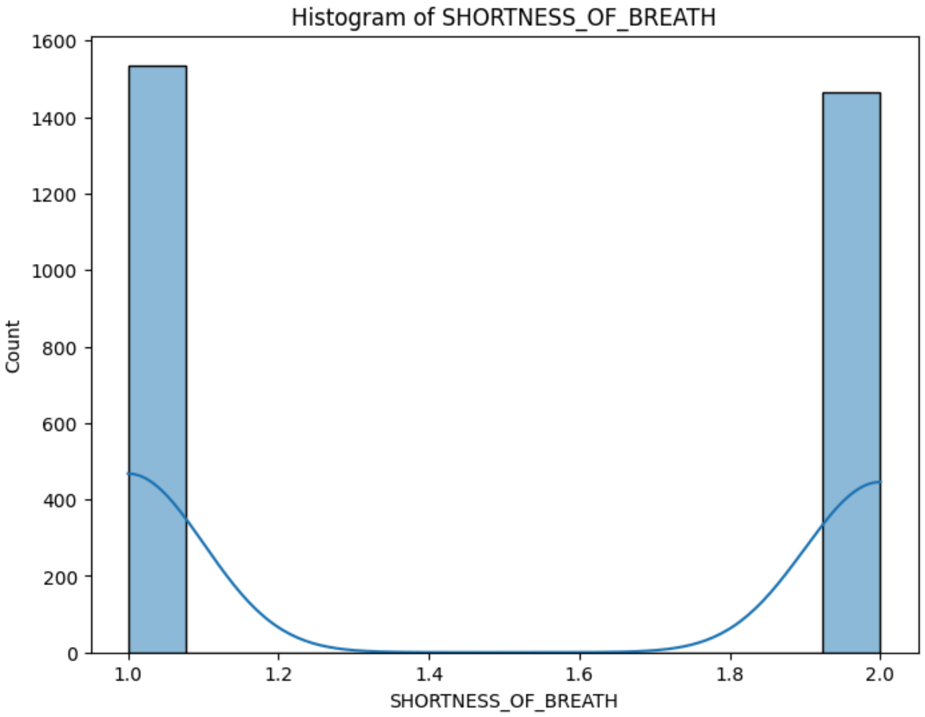

Histogram of ALLERGY
Count
ALLERGY
0
200
400
600
800
1000
1200
1400
1.0
1.2
1.4
1.6
1.8
2.0

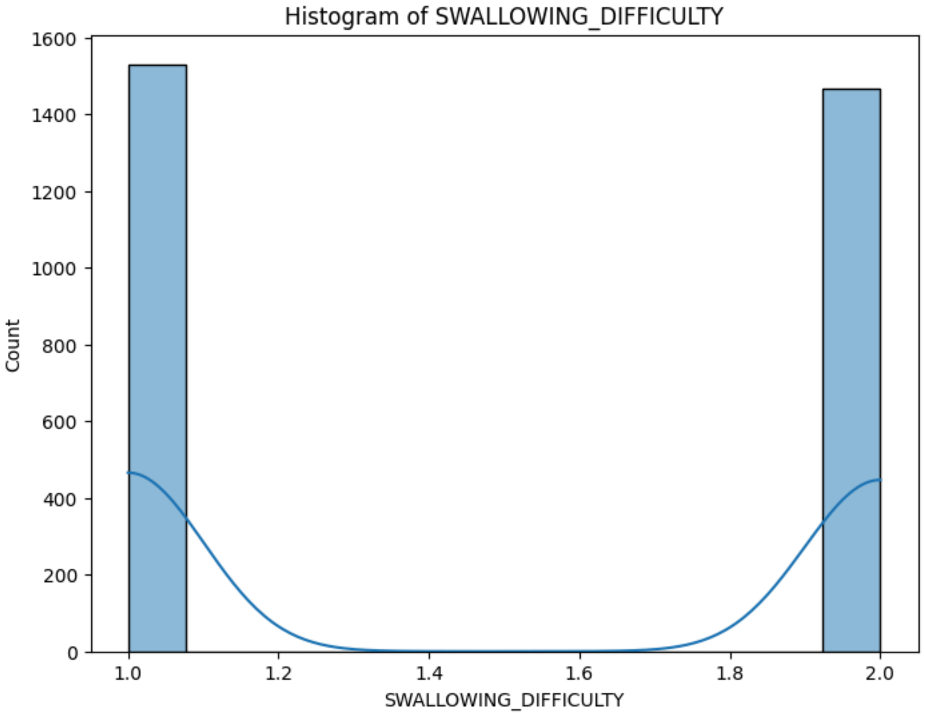

Histogram of WHEEZING
Count
WHEEZING
0
200
400
600
800
1000
1200
1400
1.0
1.2
1.4
1.6
1.8
2.0

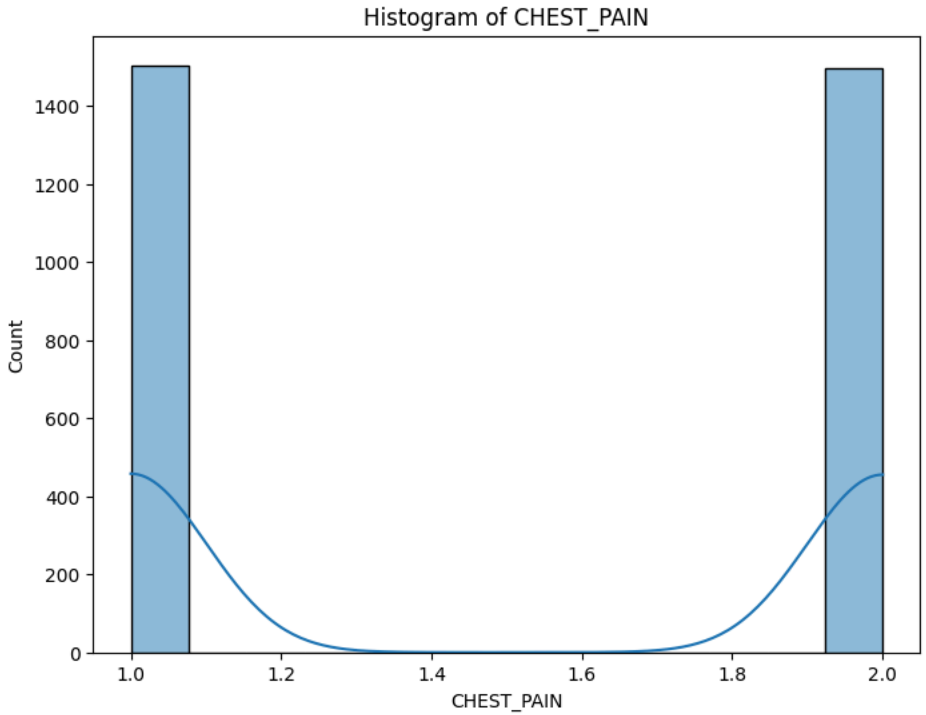


Histogram of ALCOHOL_CONSUMING
Count
ALCOHOL_CONSUMING
0
200
400
600
800
1000
1200
1400
1600
1.0
1.2
1.4
1.6
1.8
2.0

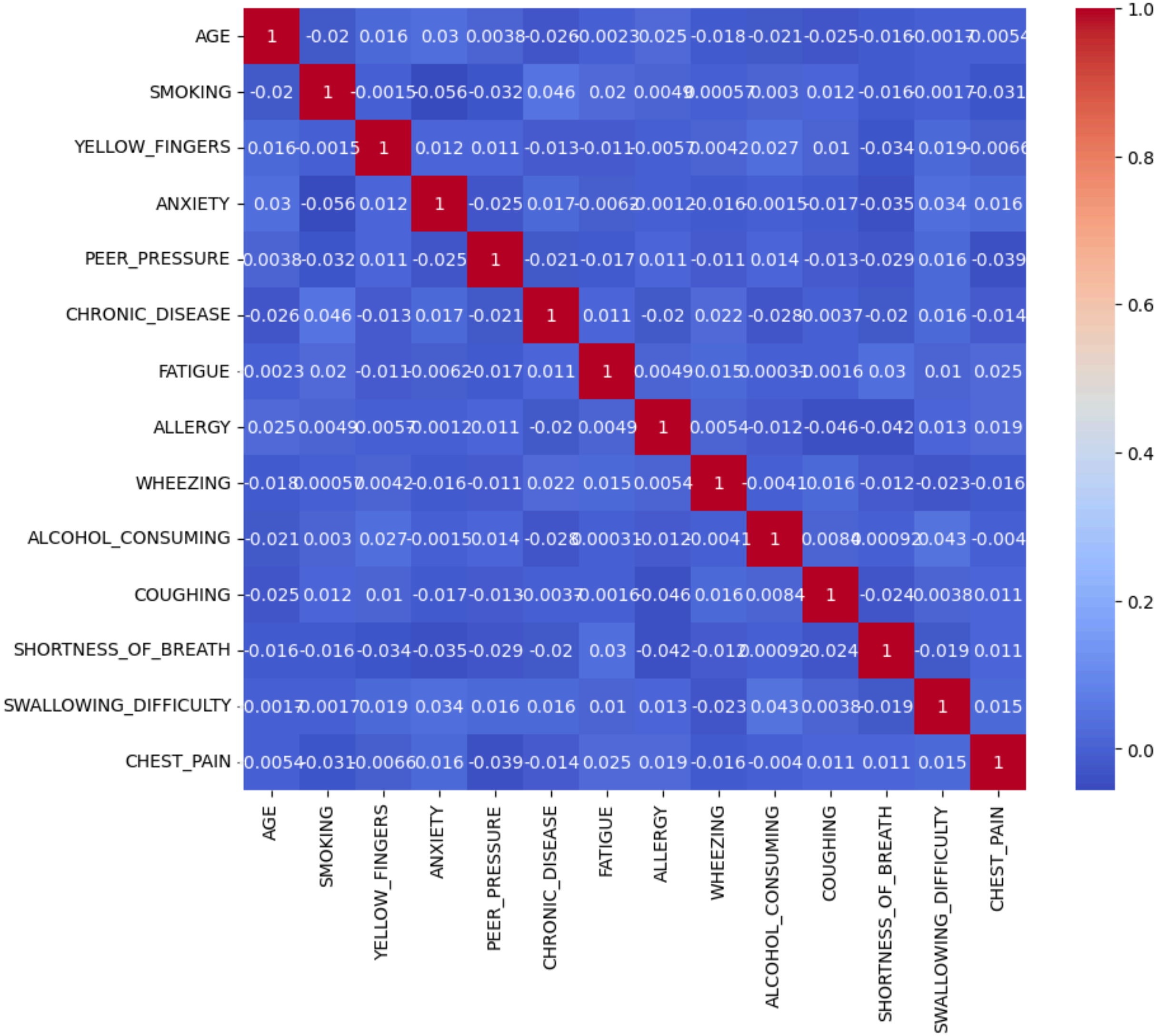

Histogram of COUGHING
1600
1400
1200
1000
800
600
400
200
0
Count
1.0
1.2
1.4
1.6
1.8
2.0
COUGHING

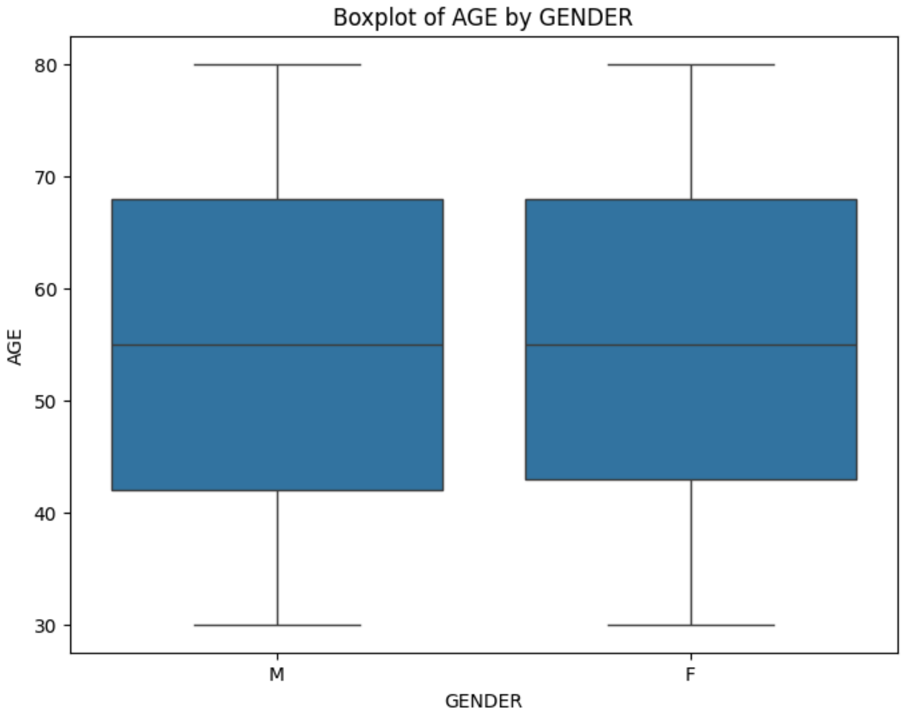

Histogram of SHORTNESS_OF_BREATH
Count
SHORTNESS_OF_BREATH
0
200
400
600
800
1000
1200
1400
1600
1.0
1.2
1.4
1.6
1.8
2.0

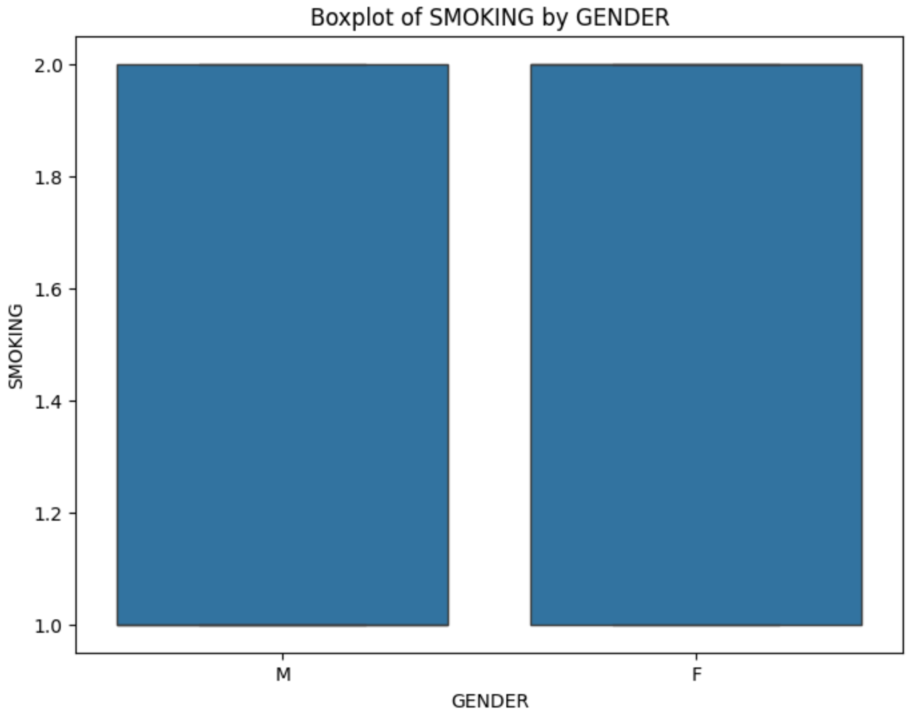

Histogram of SWALLOWING_DIFFICULTY
Count
SWALLOWING_DIFFICULTY
0
200
400
600
800
1000
1200
1400
1600
1.0
1.2
1.4
1.6
1.8
2.0

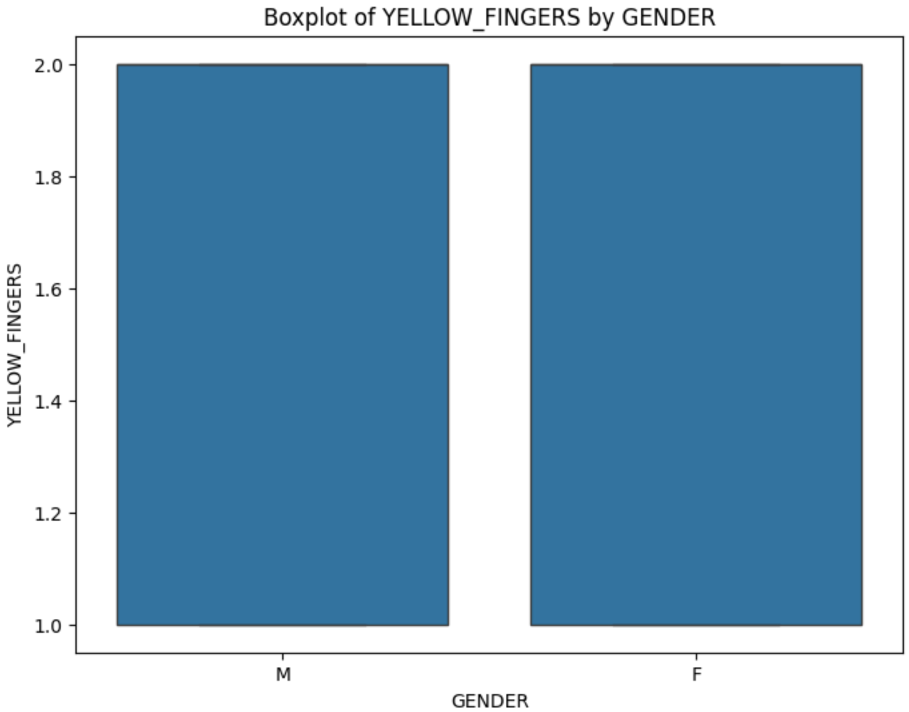


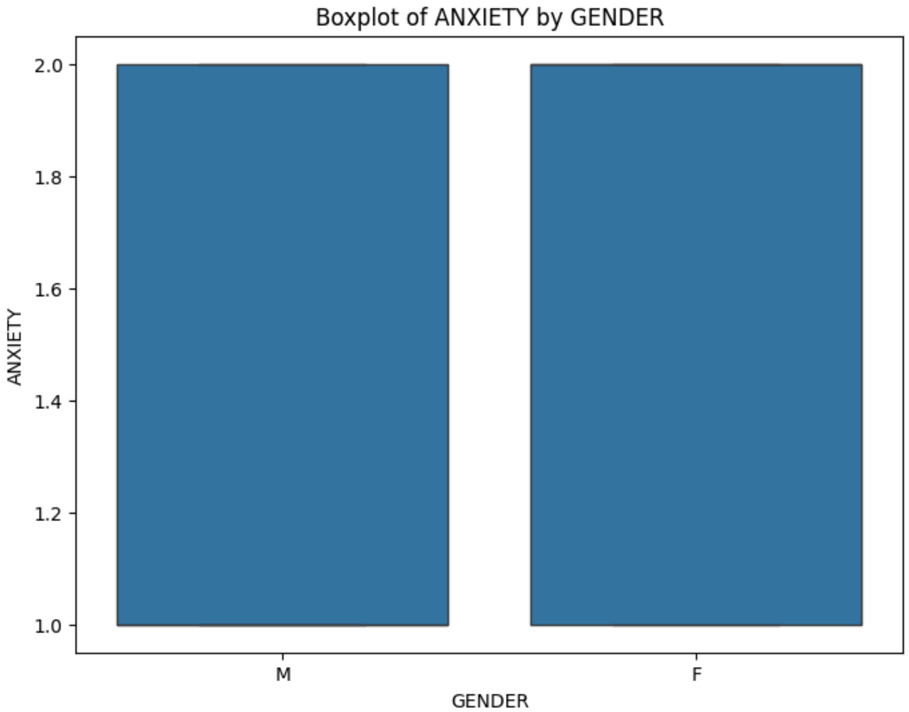

```
'Correlation between numerical variables:'
```

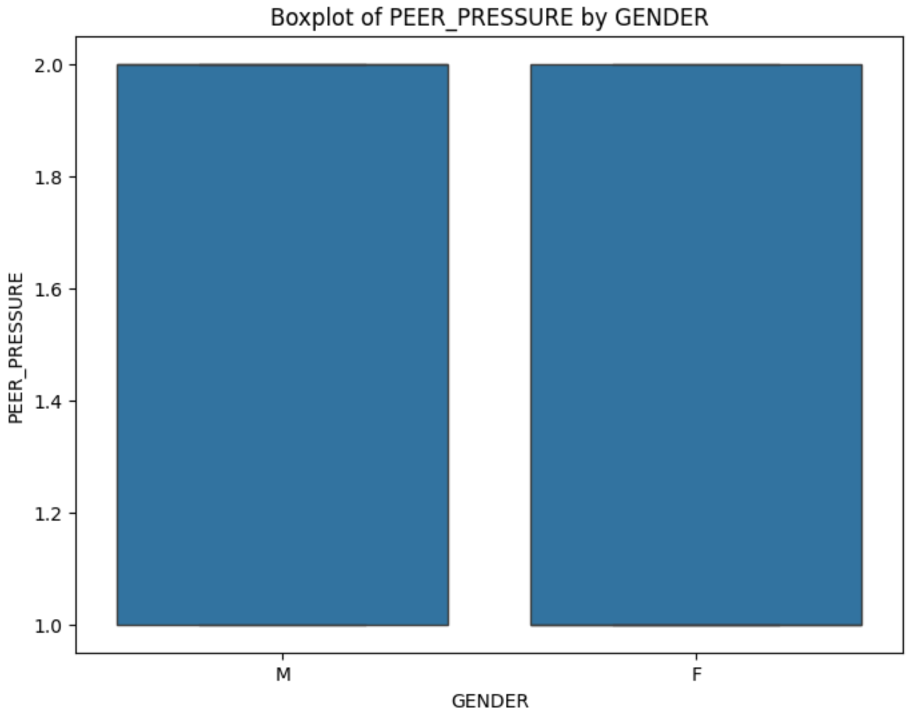

AGE
SMOKING
YELLOW_FINGERS
ANXIETY
PEER_PRESSURE
CHRONIC_DISEASE
FATIGUE
ALLERGY
WHEEZING
ALCOHOL_CONSUMING
COUGHING
SHORTNESS_OF_BREATH
SWALLOWING_DIFFICULTY
CHEST_PAIN
1.0
0.8
0.6
0.4
0.2
0.0

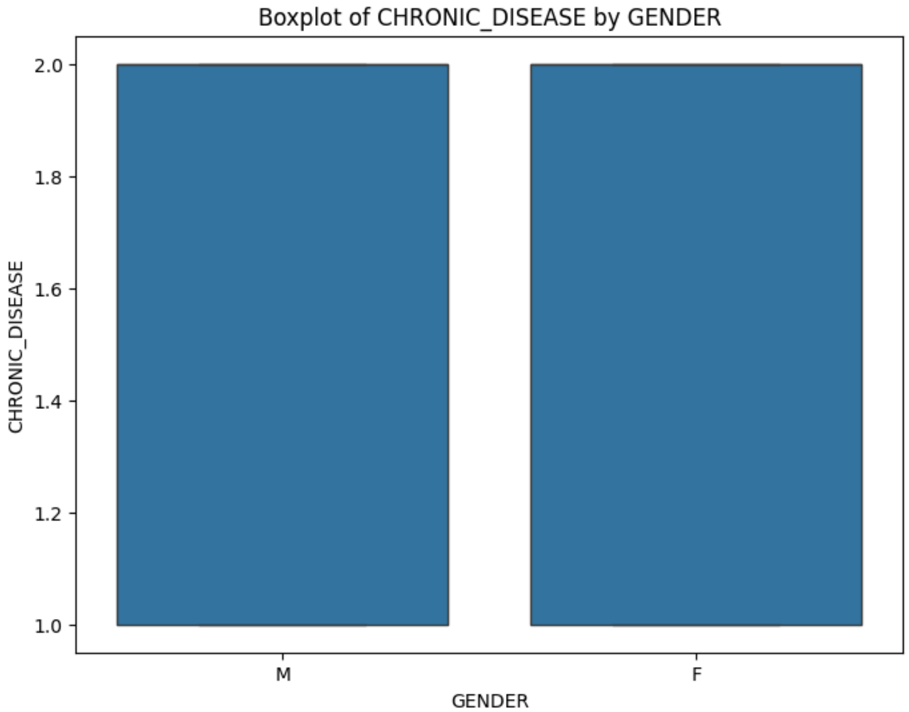

Boxplot of AGE by GENDER
80
70
60
50
40
30
AGE
M
F
GENDER

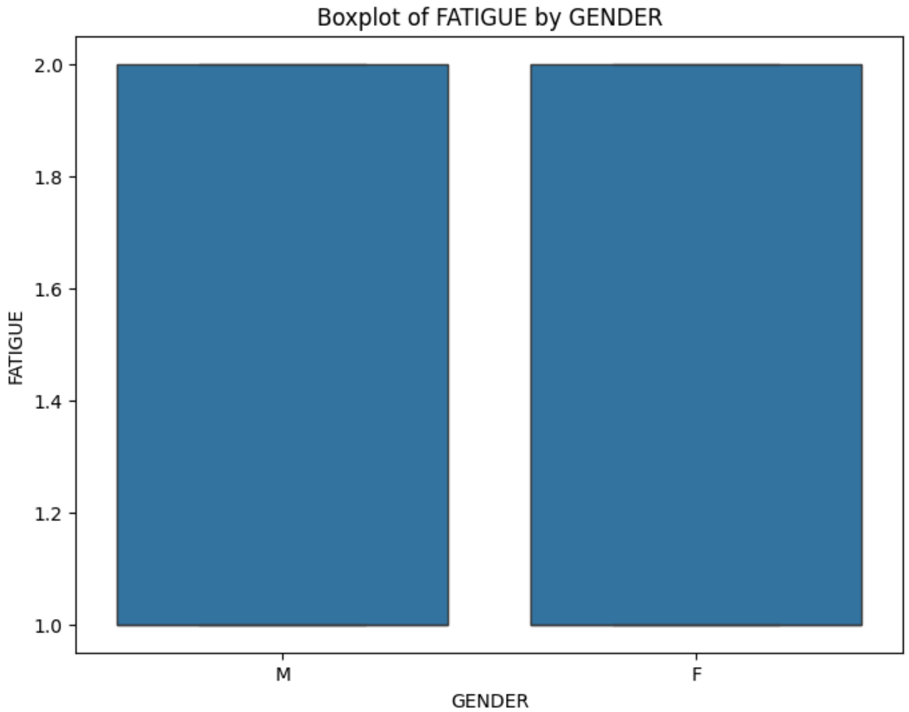

Boxplot of SMOKING by GENDER
2.0
1.8
1.6
1.4
1.2
1.0
SMOKING
M
F
GENDER

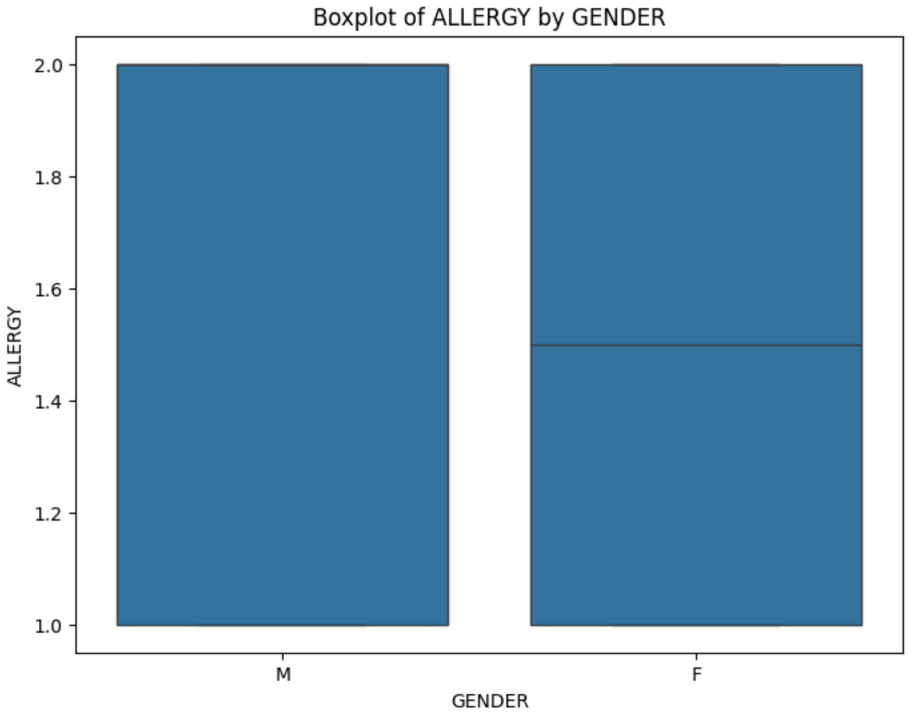

Boxplot of YELLOW_FINGERS by GENDER
2.0
1.8
1.6
1.4
1.2
1.0
YELLOW_FINGERS
M
F
GENDER

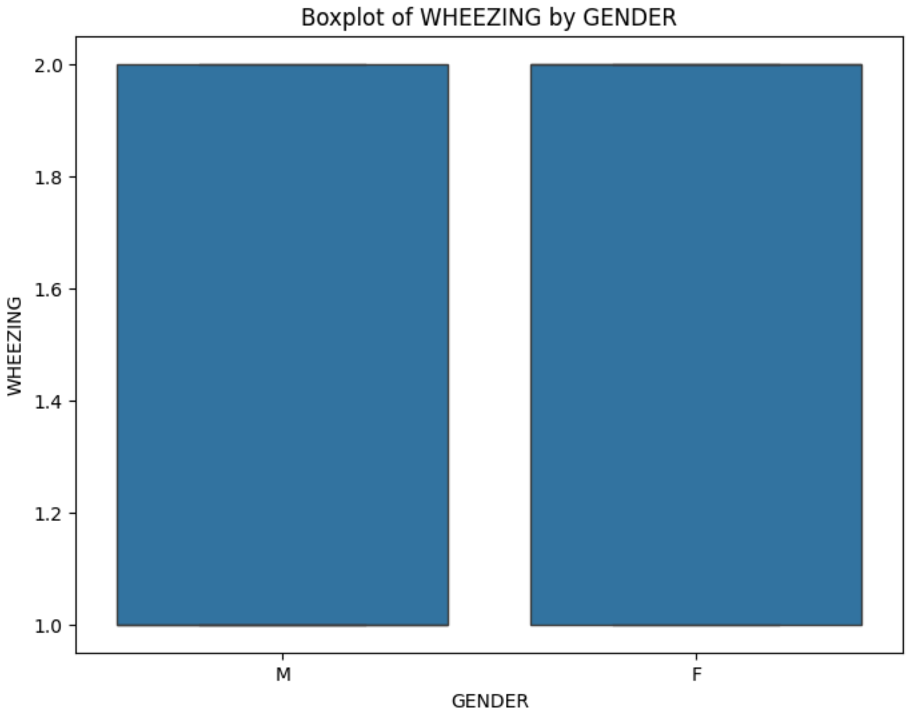

Boxplot of ANXIETY by GENDER
2.0
1.8
1.6
1.4
1.2
1.0
ANXIETY
M
F
GENDER

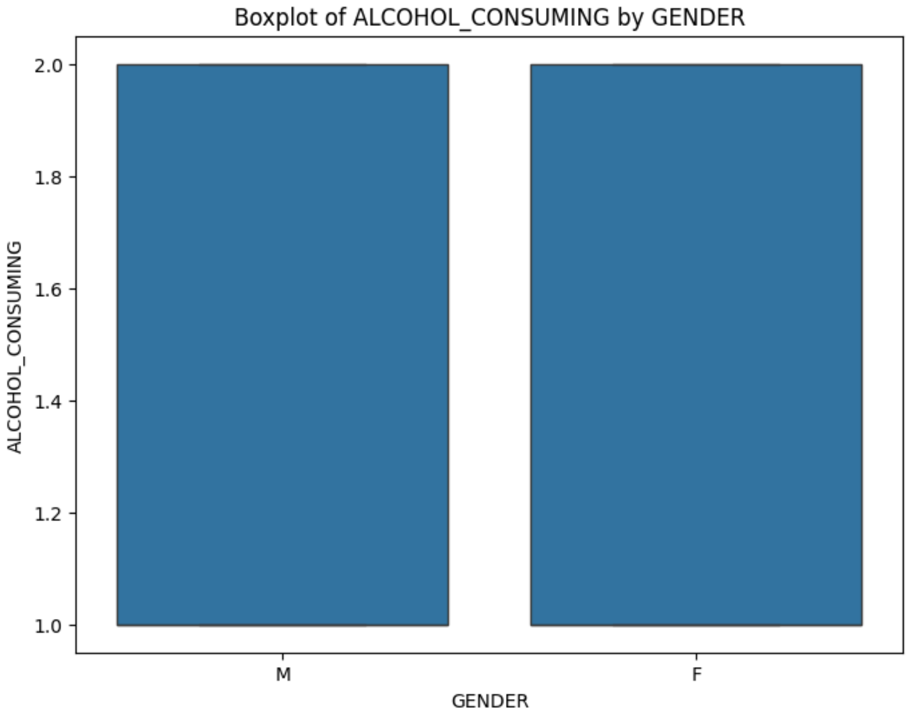

Boxplot of PEER_PRESSURE by GENDER
2.0
1.8
1.6
1.4
1.2
1.0
PEER_PRESSURE
M
F
GENDER

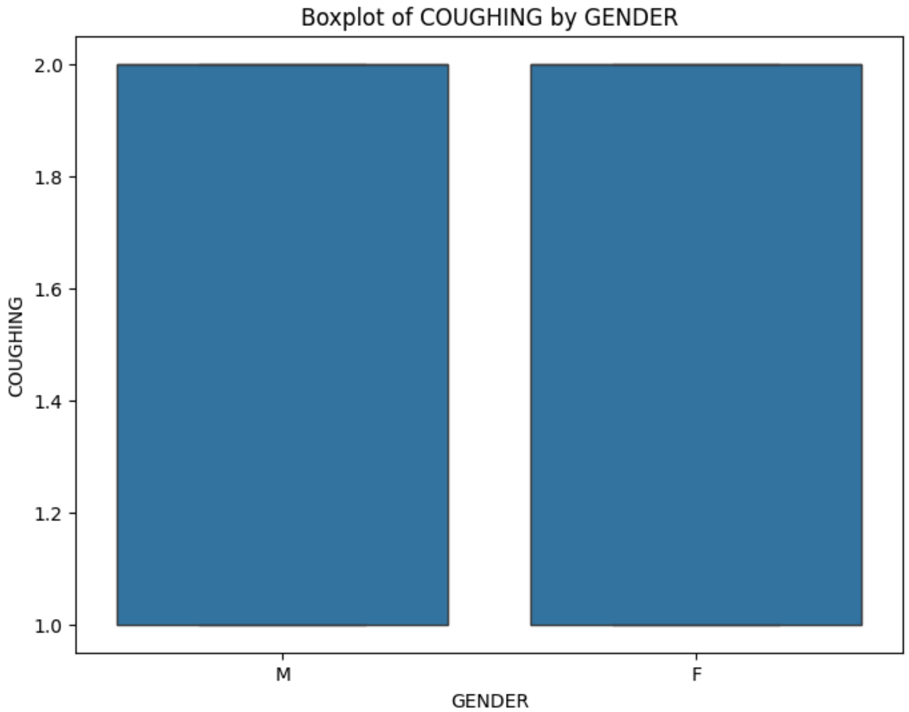

Boxplot of CHRONIC_DISEASE by GENDER
2.0
1.8
1.6
1.4
1.2
1.0
CHRONIC_DISEASE
M
F
GENDER

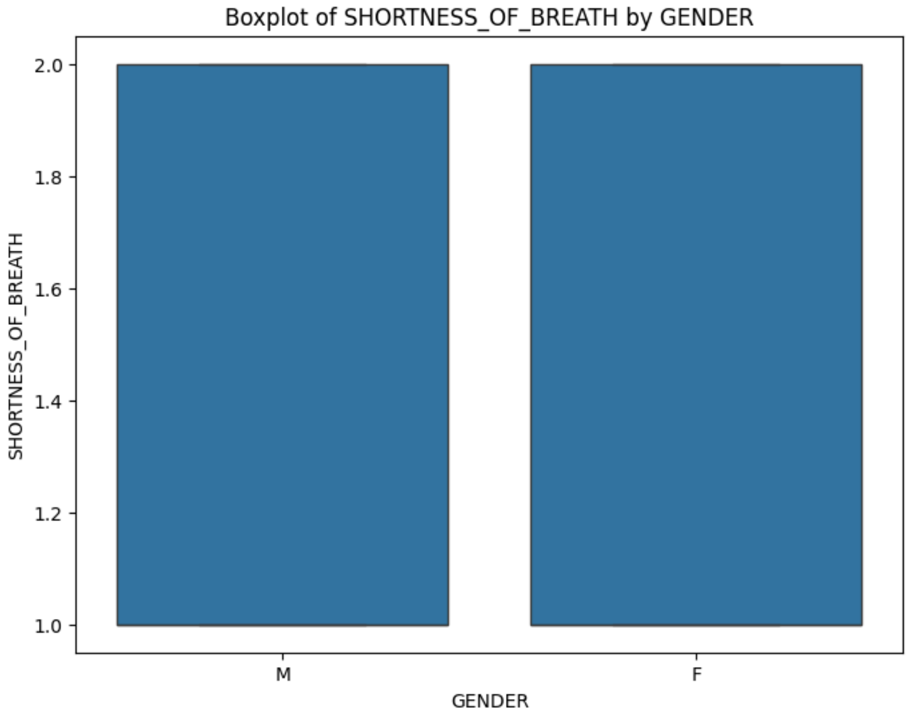

Boxplot of FATIGUE by GENDER
2.0
1.8
1.6
1.4
1.2
1.0
FATIGUE
M
F
GENDER

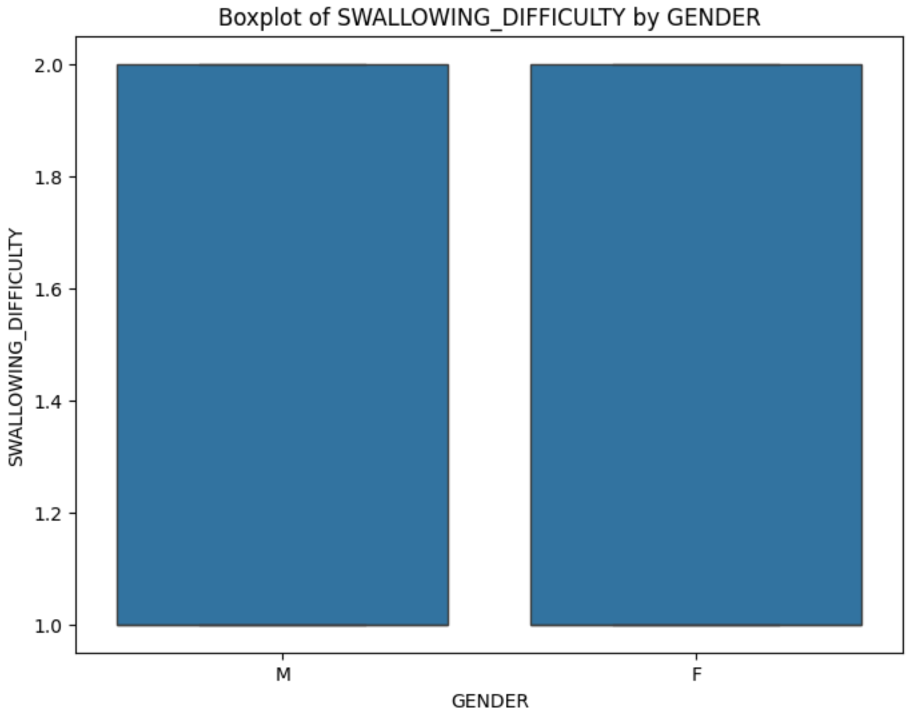

Boxplot of ALLERGY by GENDER
2.0
1.8
1.6
1.4
1.2
1.0
ALLERGY
M
F
GENDER

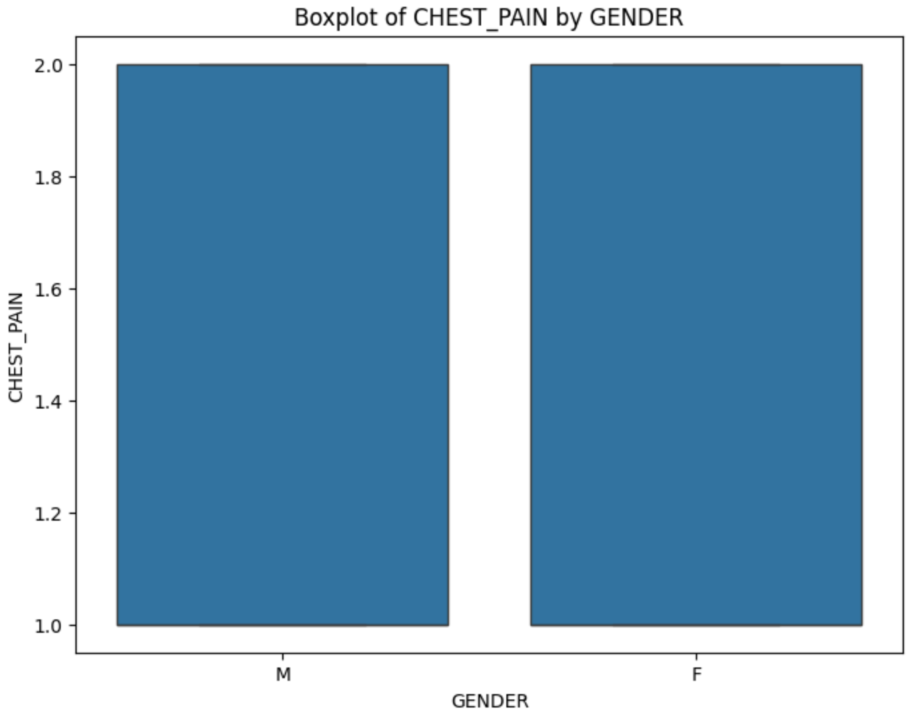

Boxplot of WHEEZING by GENDER
2.0
1.8
1.6
1.4
1.2
1.0
WHEEZING
M
F
GENDER

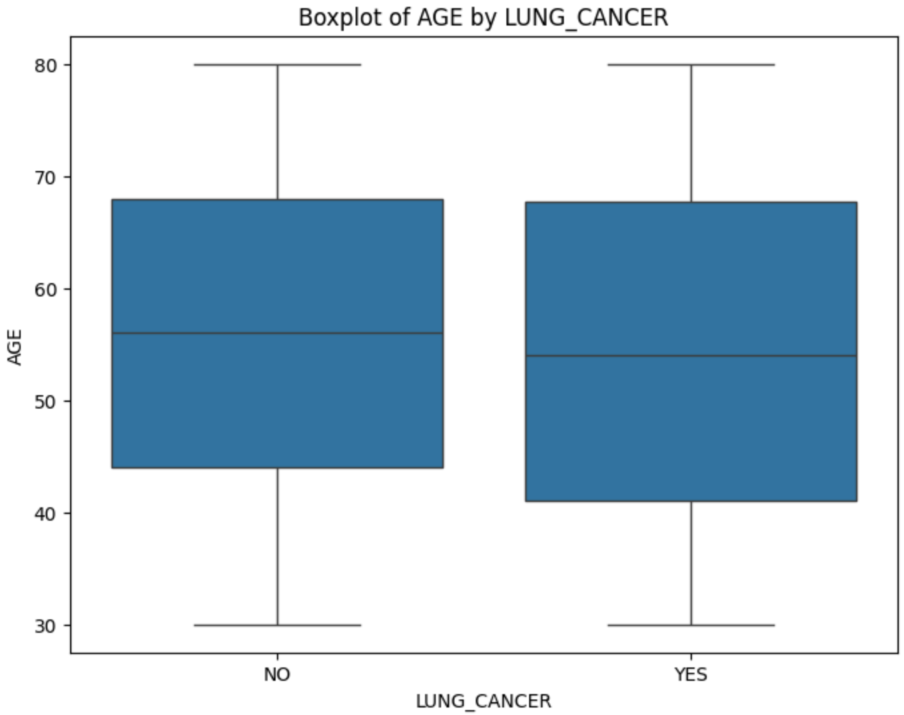

Boxplot of ALCOHOL_CONSUMING by GENDER
2.0
1.8
1.6
1.4
1.2
1.0
ALCOHOL_CONSUMING
M
F
GENDER

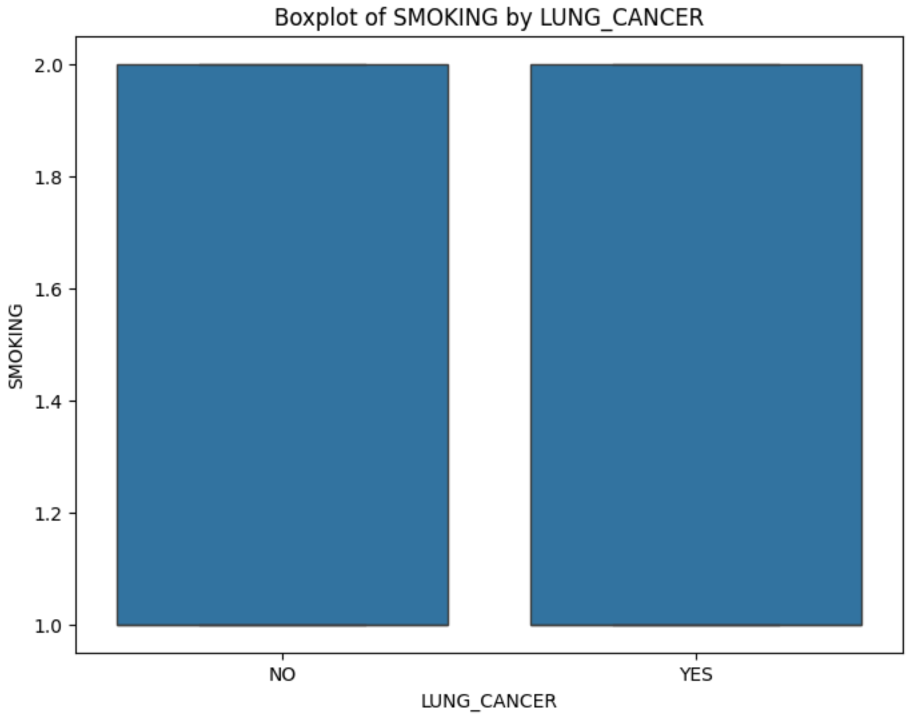

Boxplot of COUGHING by GENDER
2.0
1.8
1.6
1.4
1.2
1.0
COUGHING
M
F
GENDER

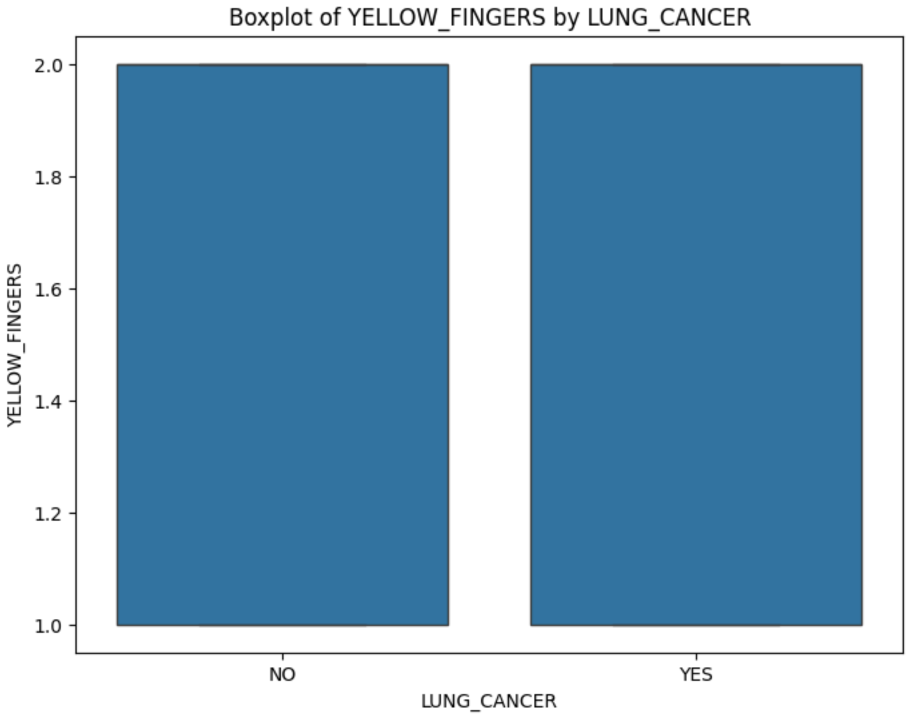

Boxplot of SHORTNESS_OF_BREATH by GENDER
2.0
1.8
1.6
1.4
1.2
1.0
SHORTNESS_OF_BREATH
M
F
GENDER

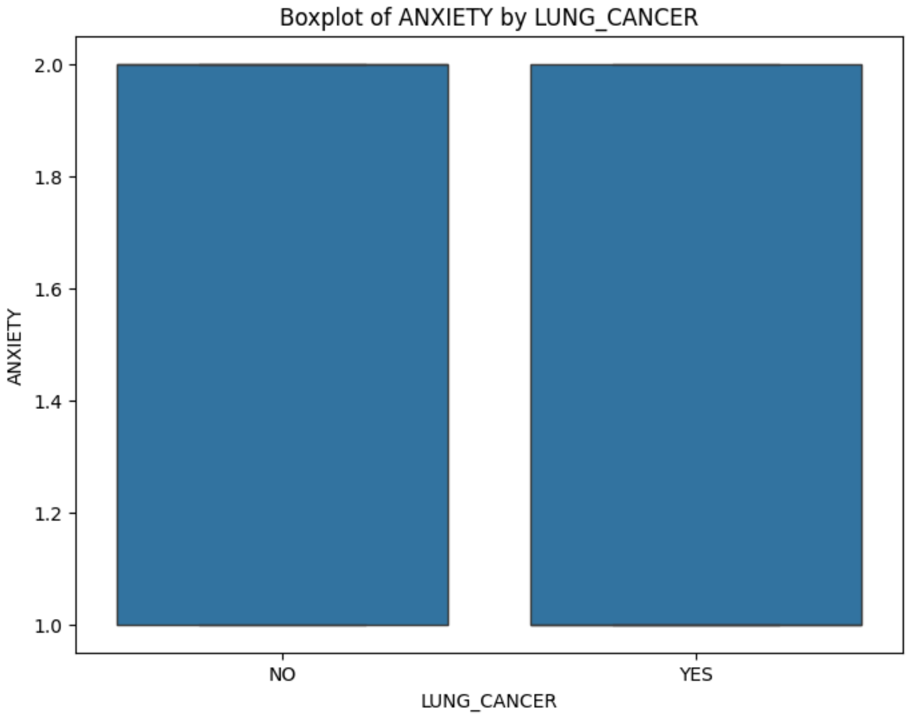

Boxplot of SWALLOWING_DIFFICULTY by GENDER
2.0
1.8
1.6
1.4
1.2
1.0
SWALLOWING_DIFFICULTY
M
F
GENDER

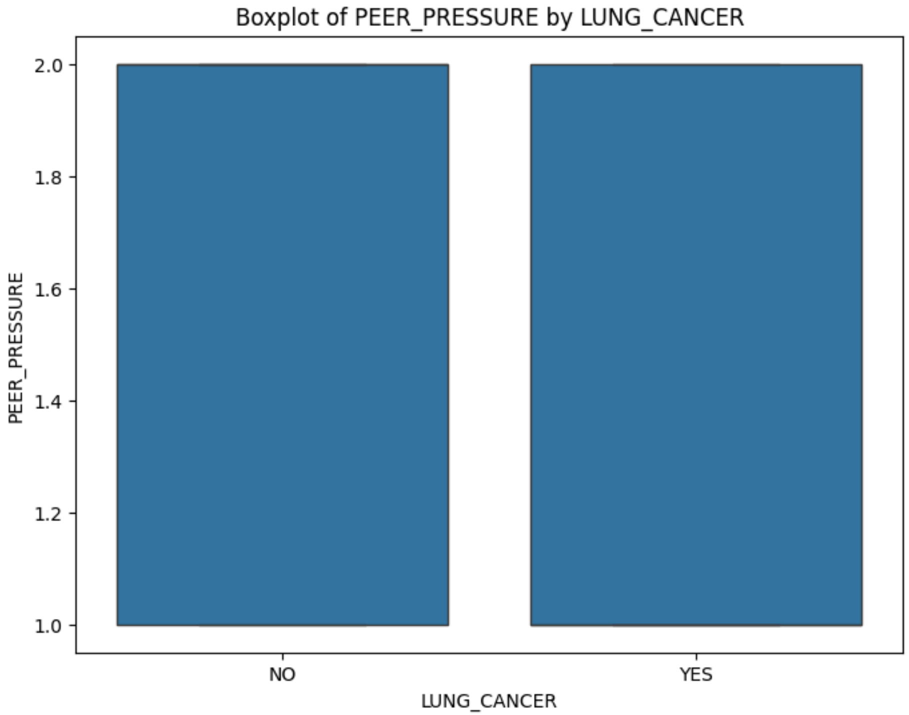

Boxplot of CHEST_PAIN by GENDER
2.0
1.8
1.6
1.4
1.2
1.0
CHEST_PAIN
M
F
GENDER

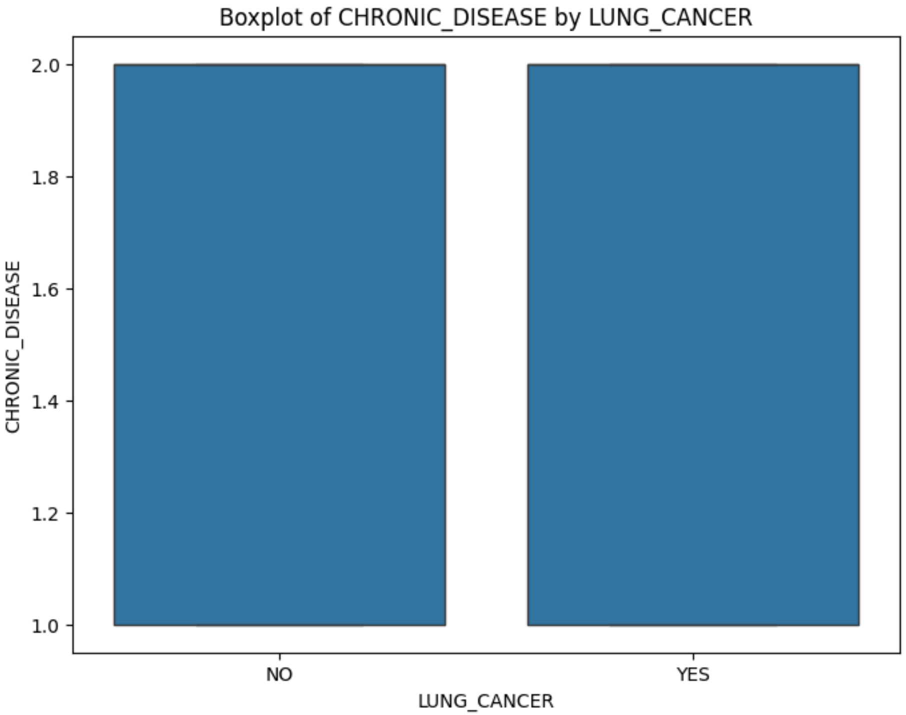

Boxplot of AGE by LUNG_CANCER
80
70
60
50
40
30
AGE
NO
YES
LUNG_CANCER

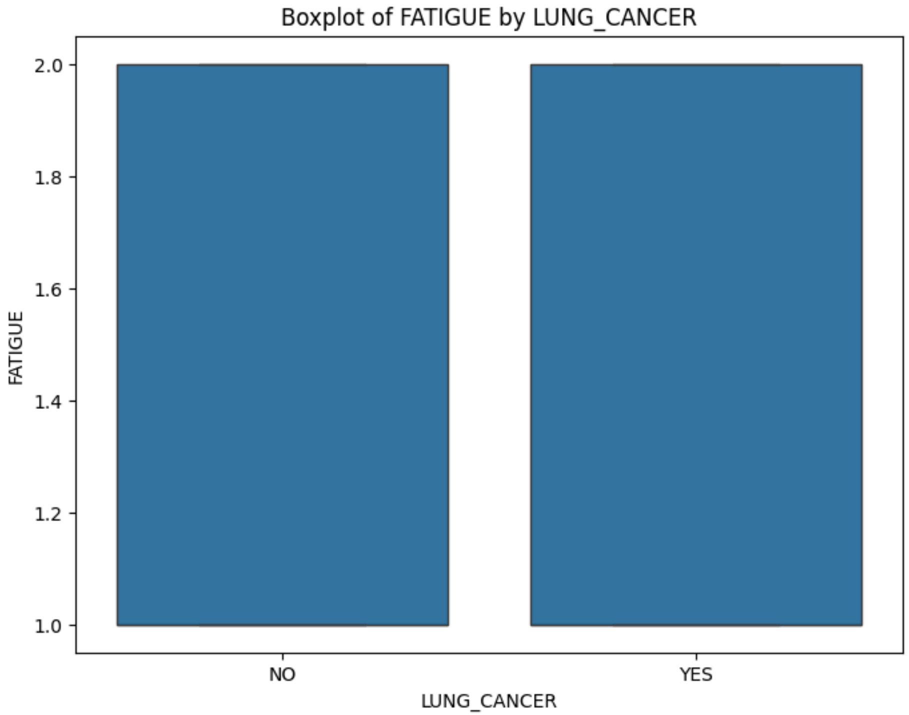

Boxplot of SMOKING by LUNG_CANCER
2.0
1.8
1.6
1.4
1.2
1.0
SMOKING
NO
YES
LUNG_CANCER

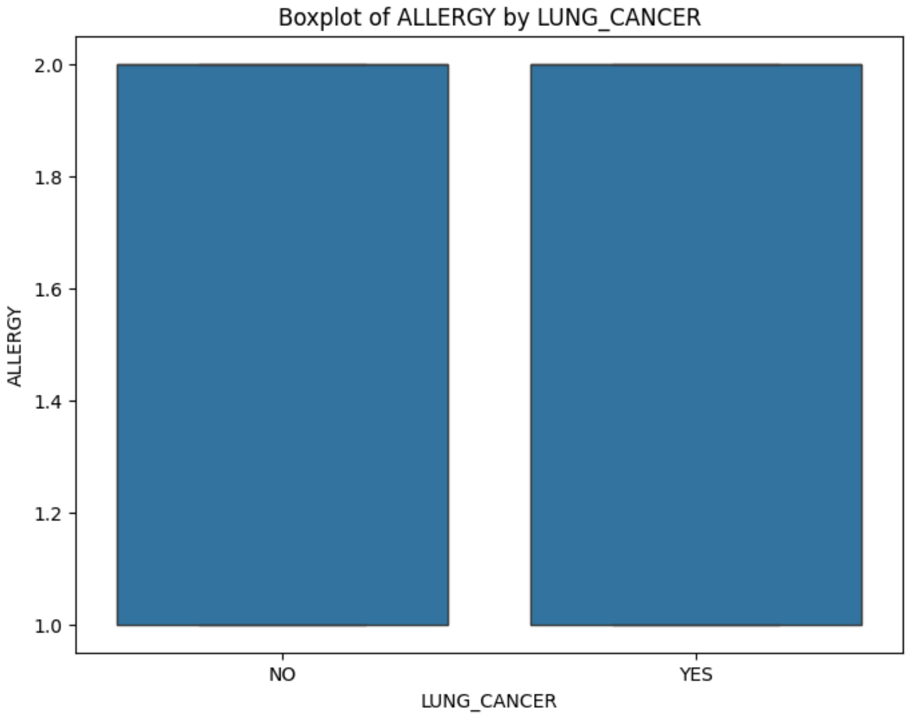

Boxplot of YELLOW_FINGERS by LUNG_CANCER
2.0
1.8
1.6
1.4
1.2
1.0
YELLOW_FINGERS
NO
YES
LUNG_CANCER

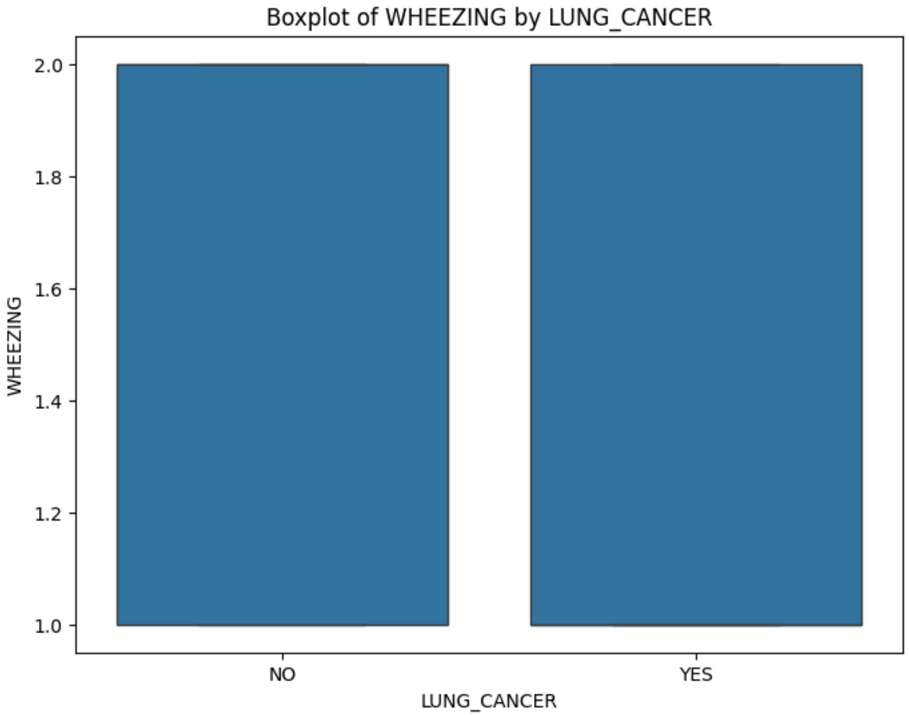

Boxplot of ANXIETY by LUNG_CANCER
2.0
1.8
1.6
1.4
1.2
1.0
ANXIETY
NO
YES
LUNG_CANCER

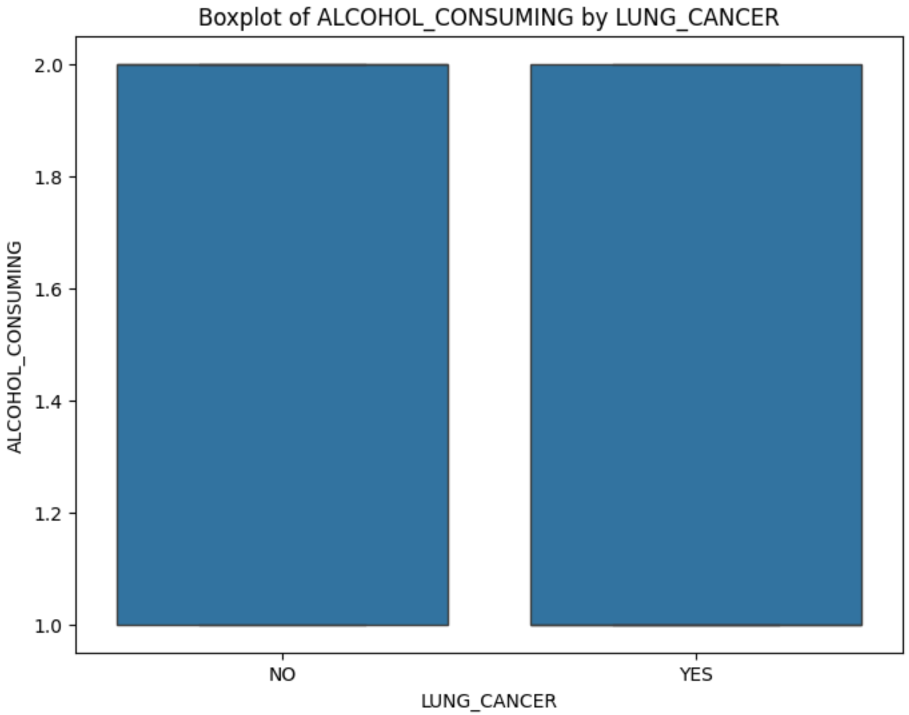

Boxplot of PEER_PRESSURE by LUNG_CANCER
2.0
1.8
1.6
1.4
1.2
1.0
PEER_PRESSURE
NO
YES
LUNG_CANCER

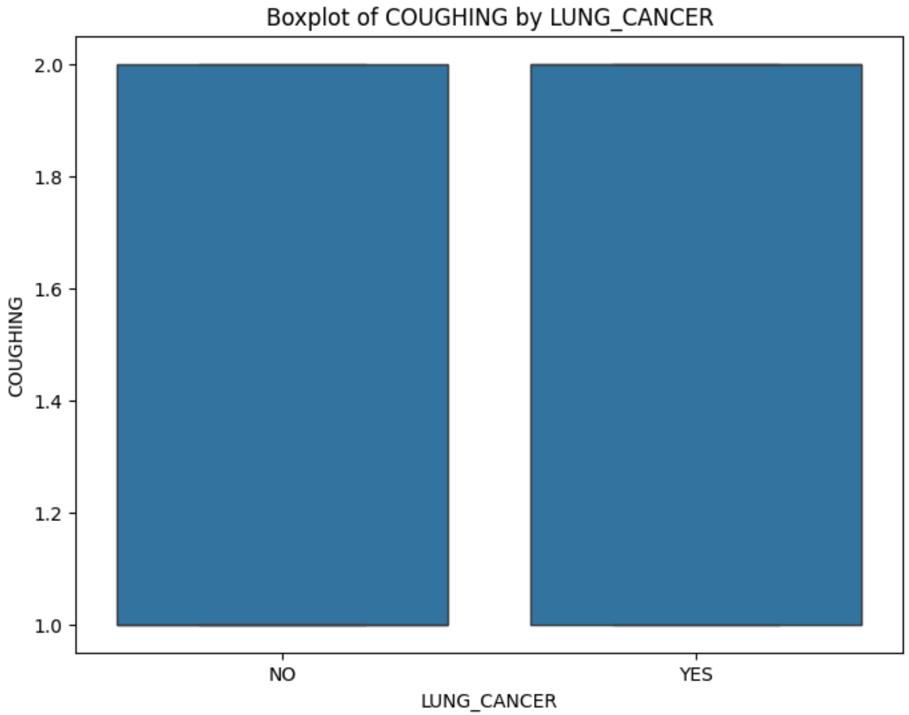

Boxplot of CHRONIC_DISEASE by LUNG_CANCER
2.0
1.8
1.6
1.4
1.2
1.0
CHRONIC_DISEASE
NO
YES
LUNG_CANCER

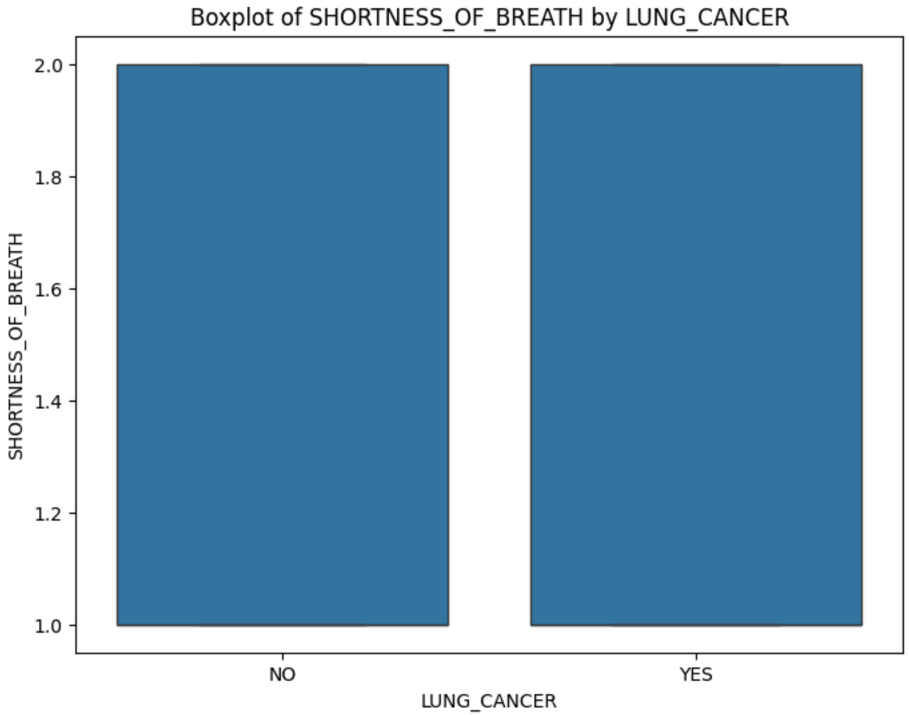

Boxplot of FATIGUE by LUNG_CANCER
2.0
1.8
1.6
1.4
1.2
1.0
FATIGUE
NO
YES
LUNG_CANCER

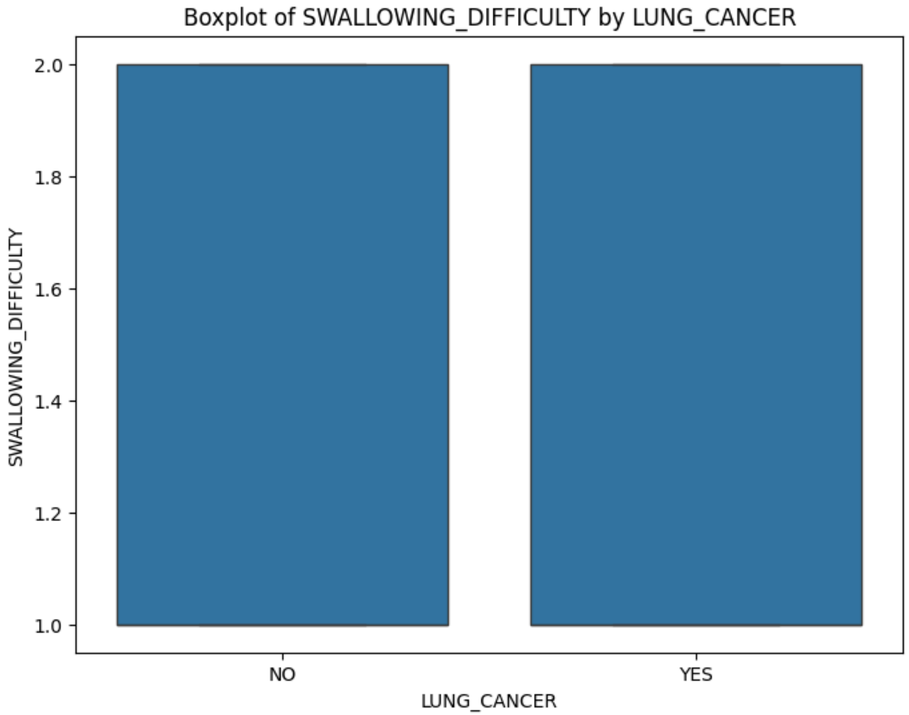

Boxplot of ALLERGY by LUNG_CANCER
2.0
1.8
1.6
1.4
1.2
1.0
ALLERGY
NO
YES
LUNG_CANCER

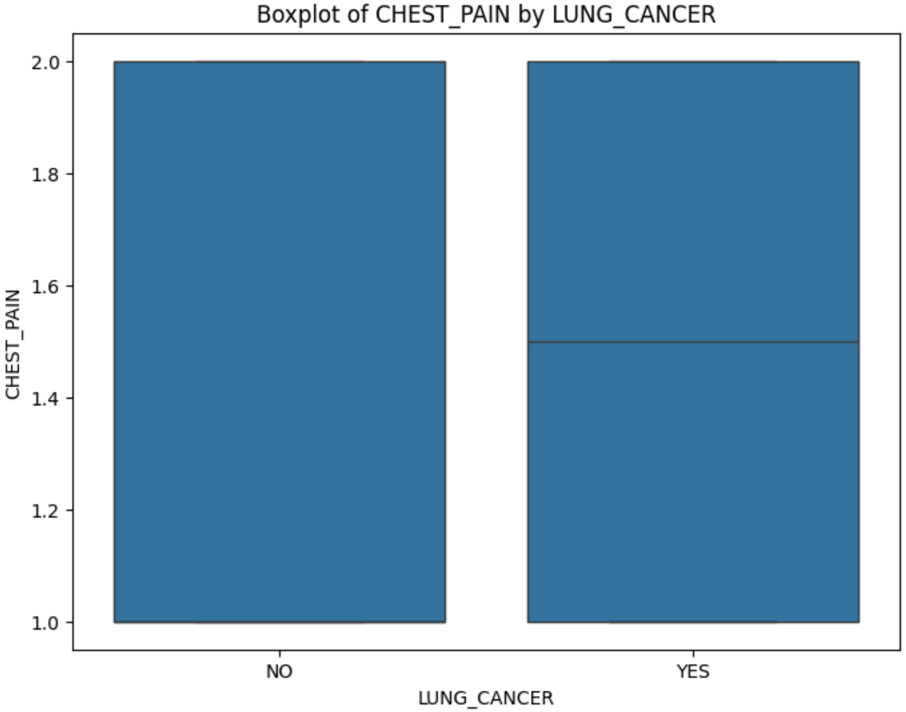

Boxplot of WHEEZING by LUNG_CANCER
2.0
1.8
1.6
1.4
1.2
1.0
WHEEZING
NO
YES
LUNG_CANCER


Boxplot of ALCOHOL_CONSUMING by LUNG_CANCER
2.0
1.8
1.6
1.4
1.2
1.0
ALCOHOL_CONSUMING
NO
YES
LUNG_CANCER


Boxplot of COUGHING by LUNG_CANCER
2.0
1.8
1.6
1.4
1.2
1.0
COUGHING
NO
YES
LUNG_CANCER


Boxplot of SHORTNESS_OF_BREATH by LUNG_CANCER
SHORTNESS_OF_BREATH
2.0
1.8
1.6
1.4
1.2
1.0
NO
YES
LUNG_CANCER


Boxplot of SWALLOWING_DIFFICULTY by LUNG_CANCER
SWALLOWING_DIFFICULTY
2.0
1.8
1.6
1.4
1.2
1.0
NO
YES
LUNG_CANCER



Boxplot of CHEST_PAIN by LUNG_CANCER
2.0
1.8
1.6
1.4
1.2
1.0
CHEST_PAIN
NO
YES
LUNG_CANCER